

Generation Interconnection Request System Impact Study Report

For Queue Project Q22-2247

Texas Farms Tap 138 kV
September 03, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2247. The Interconnection Customer proposes a 737.9 MW natural gas combined cycle facility interconnecting at the Texas Farms Tap 138 kV substation in the West Virginia area. The request was received on December 04, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 527 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 23 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$185,658,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Eastshore – Amboy 138 kV
Contingency: Radisson – Texas Farms Tap 138 kV L
Loading: 287.1% (Rating: 152 MVA)
DFAX: 55.7%
Pre-project loading: 17.1%
2. Facility: North Adams – Radisson 138 kV
Contingency: Amboy Substation – Texas Farms Tap 138 kV
Loading: 265.5% (Rating: 193 MVA)
DFAX: 62.0%
Pre-project loading: 28.2%
3. Facility: Texas Farms Tap – Radisson 138 kV
Contingency: Texas Farms Tap – Amboy 138 kV line
Loading: 255.6% (Rating: 202 MVA)
DFAX: 62.0%
Pre-project loading: 29.2%
4. Facility: Dillard-Orange Substation – Eastshore Substation 138 kV
Contingency: Texas Farms Tap – Radisson 138 kV
Loading: 255.5% (Rating: 170 MVA)
DFAX: 55.7%
Pre-project loading: 13.6%
5. Facility: Texas Farms Tap – Amboy 138 kV
Contingency: Radisson – Texas Farms Tap 138 kV
Loading: 234.0% (Rating: 191 MVA)
DFAX: 55.7%
Pre-project loading: 18.4%

6. Facility: Waterton – Iron Grove 138 kV
Contingency: Mattoon E – Rockaway Beach 138 kV
Loading: 159.3% (Rating: 220 MVA)
DFAX: 7.2%
Pre-project loading: 135.1%
7. Facility: Panda – Goat Rock 138 kV
Contingency: Panda – Northern Steel Casting 138 kV
Loading: 158.8% (Rating: 217 MVA)
DFAX: 5.6%
Pre-project loading: 139.8%
8. Facility: Northern Steel Casting – Goat Rock Substation 138 kV line
Contingency: Northern Steel Casting – Panda 138 kV
Loading: 152.7% (Rating: 229 MVA)
DFAX: 5.6%
Pre-project loading: 134.6%
9. Facility: Waterton – Hampshire Paper 138 kV
Contingency: Mattoon E Substation – Rockaway Beach Substation 138 kV
Loading: 150.9% (Rating: 227 MVA)
DFAX: 7.2%
Pre-project loading: 127.5%
10. Facility: Panda – Northern Steel Casting 138 kV
Contingency: Fishkill Plains – Dillard-Orange 345 kV
Loading: 142.4% (Rating: 243 MVA)
DFAX: 5.4%
Pre-project loading: 126.0%
11. Facility: Union Valley – Iron Grove 138 kV L
Contingency: Radisson – Texas Farms Tap 138 kV
Loading: 137.9% (Rating: 204 MVA)
DFAX: 23.4%
Pre-project loading: 53.5%
12. Facility: East Sidney – Whitefish Substation 138 kV
Contingency: Fishkill Plains – Dillard-Orange 345 kV Line
Loading: 134.6% (Rating: 267 MVA)
DFAX: 12.0%
Pre-project loading: 101.5%
13. Facility: East Sidney – Semiahmoo 138 kV
Contingency: Fishkill Plains – Dillard-Orange 345 kV
Loading: 134.1% (Rating: 267 MVA)
DFAX: 12.0%
Pre-project loading: 101.0%
14. Facility: Lemon Gulch – Texas Farms Tap 138 kV Line
Contingency: Txas Farms Tap – Radisson 138 kV
Loading: 130.0% (Rating: 221 MVA)
DFAX: 44.3%
Pre-project loading: 18.0%
15. Facility: Grana 1907-1909 – Huston 138 kV line
Contingency: Fishkill Plains – Dillard-Orange 345 kV line
Loading: 126.8% (Rating: 285 MVA)
DFAX: 9.2%
Pre-project loading: 102.9%
16. Facility: Grana 1907-1909 – Pana N 138 kV
Contingency: Dillard-Orange – Fishkill Plains 345 kV L

Loading: 126.3% (Rating: 294 MVA)

DFAX: 9.2%

Pre-project loading: 103.1%

17. Facility: Aguada 7201 – Lemon Gulch Substation 138 kV line

Contingency: Texas Farms Tap – Radisson 138 kV

Loading: 126.0% (Rating: 225 MVA)

DFAX: 44.3%

Pre-project loading: 19.2%

18. Facility: Rec Harmony – Pana Norh 138 kV

Contingency: Dillard-Orange – Fishkill Plains 345 kV

Loading: 125.8% (Rating: 302 MVA)

DFAX: 9.2%

Pre-project loading: 103.3%

19. Facility: Rec Hamony – N Adams 138 kV Line

Contingency: Texas Farms Tap – Amboy 138 kV

Loading: 124.4% (Rating: 365 MVA)

DFAX: 50.0%

Pre-project loading: 23.3%

20. Facility: Union Valley – Aguada7201 138 kV

Contingency: Radisson – Texas Farms Tap 138 kV line

Loading: 120.1% (Rating: 230 MVA)

DFAX: 44.3%

Pre-project loading: 22.1%

21. Facility: Panda – Menlo 138 kV L

Contingency: Fishkill Plains – Dillard-Orange 345 kV L

Loading: 116.3% (Rating: 383 MVA)

DFAX: 7.1%

Pre-project loading: 102.7%

22. Facility: Solana – Northern Steel Csting 138 kV

Contingency: FishkillPlains – Dillrd-Orange 345 kV

Loading: 115.6% (Rating: 397 MVA)

DFAX: 7.1%

Pre-project loading: 102.5%

23. Facility: Solana – Hampshire Paper 138 kV line

Contingency: Fishkill Plains – Dillard-Orange Substation 345 kV Line

Loading: 115.4% (Rating: 403 MVA)

DFAX: 7.1%

Pre-project loading: 102.4%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Easshore – Amboy 138 kV	Rebuild line with higher-capacity conductor	\$3,867,000	100% (\$3,867,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
North Adams – Radisson 138 kV	Rebuild line with higher-capacity conductor	\$12,762,000	100% (\$12,762,000)
Rdisson – Texas Farms Tap 138 kV	Rebuild line with higher-capacity conductor	\$7,701,000	100% (\$7,701,000)
Dillard-Orange – Eastshore 138 kV	Rebuild line with higher-capacity conductor	\$16,656,000	100% (\$16,656,000)
Texas Farms Tap – Amboy Substation 138 kV L	Rebuild line with higher-capacity conductor	\$6,843,000	100% (\$6,843,000)
Waterton – Iron Grove 138 kV	Rebuild line with higher-capacity conductor	\$5,569,000	100% (\$5,569,000)
Goat Rock – Panda 138 kV L	Rebuild line with higher-capacity conductor	\$8,367,000	100% (\$8,367,000)
Northern Steel Casting – Goat Rock Substation 138 kV line	Rebuild line with higher-capacity conductor	\$5,397,000	100% (\$5,397,000)
Waterton – Hampshire Paper 138 kV	Rebuild line with higher-capacity conductor	\$5,834,000	100% (\$5,834,000)
Panda – Northern Steel Casting 138 kV	Rebuild line with higher-capacity conductor	\$3,266,000	100% (\$3,266,000)
Union Vly – Iron Grove 138 kV	Rebuild line with higher-capacity conductor	\$4,831,000	100% (\$4,831,000)
Whitfish – East Sidney 138 kV	Rebuild line with higher-capacity conductor	\$9,330,000	95% (\$8,823,000)
Semiahmoo Substation – East Sidney 138 kV	Rebuild line with higher-capacity conductor	\$11,099,000	95% (\$10,496,000)
Lemon Gulch Substation – Texas Farms Tap 138 kV	Rebuild line with higher-capacity conductor	\$7,998,000	100% (\$7,998,000)
Grana 1907-1909 – Huston 138 kV	Rebuild line with higher-capacity conductor	\$14,937,000	100% (\$14,937,000)
Grana 1907-1909 – Pana N 138 kV	Rebuild line with higher-capacity conductor	\$3,660,000	100% (\$3,660,000)
Aguada 7201 – Lemon Gulch Substation 138 kV line	Rebuild line with higher-capacity conductor	\$2,338,000	100% (\$2,338,000)
Rec Harmony – Pana Norh 138 kV	Rebuild line with higher-capacity conductor	\$5,002,000	100% (\$5,002,000)
Rec Harmony – N Adams 138 kV Line	Rebuild line with higher-capacity conductor	\$4,159,000	100% (\$4,159,000)
Aguada 7201 – Union Valley 138 kV	Rebuild line with higher-capacity conductor	\$26,825,000	100% (\$26,825,000)
Panda – Menlo Substation 138 kV	Rebuild line with higher-capacity conductor	\$8,815,000	76% (\$6,745,000)
Solana – Northern Steel Csting 138 kV	Rebuild line with higher-capacity conductor	\$8,479,000	76% (\$6,488,000)
Solana – Hampshire Paper 138 kV line	Rebuild line with higher-capacity conductor	\$9,270,000	76% (\$7,094,000)
	Total allocated to project		\$185,658,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.